

Notes: Grossman and Helpman (JPE, 1995)

Trade Wars and Trade Talks

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1 Big picture

- **Question.** How do domestic special-interest politics shape trade wars and trade negotiations between large countries?
- **Central idea.** International trade policy is a two-level game. A government bargains with domestic lobbies at home and with foreign governments abroad. The trade policy that emerges is therefore not simply the outcome of a benevolent planner maximizing national welfare.
- **Domestic politics.** Organized industry lobbies offer campaign-contribution schedules to politicians. Politicians care about both aggregate voter welfare and political contributions. The parameter a measures how much a government weights voter welfare relative to political contributions.
- **International politics.** Governments are large enough that their trade taxes influence world prices. In a trade war, governments choose tariffs and export taxes noncooperatively. In trade talks, they bargain over policy schedules and possibly transfers.
- **Trade-war result.** Noncooperative policy contains two components: a political-support component and a terms-of-trade component. Organized import-competing industries tend to obtain higher tariffs, while unorganized export sectors tend to face export taxes.
- **Trade-talks result.** Efficient bargaining eliminates pure terms-of-trade motives. The negotiated policy wedge reflects relative political pressure at home versus abroad, not the countries' market power in trade.
- **Political-paradox result.** A government that cares less about aggregate welfare can sometimes improve its country's terms of trade because it can credibly take a more aggressive protectionist stance.
- **Negotiation result.** The distribution of gains from cooperation depends on the governments' noncooperative fallback payoffs and their patience in bargaining.
- **Empirical implication.** Protection should depend not only on domestic industry political strength but also on foreign industry political strength when policies are negotiated internationally.
- **Economic takeaway.** Trade agreements are not only solutions to external terms-of-trade inefficiencies. They are also political bargains that pit domestic pressure groups in different countries against each other.

2 Incremental contribution

- **Extends endogenous trade-policy models to large-country interaction.** Earlier endogenous protection models often studied a small country facing fixed world prices. This paper embeds the Grossman–Helpman campaign-contribution framework in a two-country model where trade policies affect world prices.
- **Combines domestic lobbying and international strategic interaction.** The paper formalizes Putnam's “two-level game” logic in a tractable economic model: governments are agents of domestic political pressures and strategic players against foreign governments.
- **Separates two motives for protection.** In a trade war, a tariff can arise from special-interest politics or from a terms-of-trade motive. The paper gives a formula that cleanly decomposes these components.
- **Shows how negotiations neutralize terms-of-trade motives but not politics.** Cooperative bargaining removes inefficient manipulation of world prices, but it does not eliminate special-interest influence. Instead, it converts protection into a contest of relative political strength across countries.
- **Introduces foreign political variables into predictions for domestic protection.** Protection in a negotiated agreement should depend on political organization, output, import/export exposure, and policy sensitivity abroad as well as at home.

- **Explains why trade talks can be desirable to self-interested politicians.** Even politicians who tolerate domestic deadweight losses for campaign contributions may dislike noncooperative external political costs imposed by foreign governments, creating a motive for international agreements.
- **Connects trade-war and trade-talk outcomes.** The paper shows how the trade-war equilibrium becomes the disagreement point in a bargaining model, linking noncooperative policy to the distribution of negotiated surplus.
- **Offers a political reinterpretation of trade agreements.** Trade agreements are not merely welfare-improving liberalization devices; they are mechanisms through which governments exchange market access while managing domestic interest-group pressures.

3 Model objects and measurement

3.1 Countries, goods, and preferences

There are two countries: home and foreign. Residents consume a numeraire good Z and n nonnumeraire goods X_i . Home preferences are additively separable:

$$U = c_Z + \sum_{i=1}^n u_i(c_{x_i}).$$

Indirect utility takes the quasilinear form

$$v(p, E) = E + S(p),$$

where $S(p)$ is consumer surplus from the nonnumeraire goods. The numeraire pins down the wage at one.

Interpretation: Quasilinear utility makes aggregate welfare expressions tractable. Since income effects do not distort demands for nonnumeraire goods, political and terms-of-trade effects can be isolated cleanly.

3.2 Production and specific factors

The numeraire uses labor only. Each nonnumeraire good uses labor and a sector-specific factor. The rent to the specific factor in industry i is $\Pi_i(p_i)$, and supply is

$$X_i(p_i) = \Pi'_i(p_i).$$

Specific-factor owners are the potential organized interests. They care about trade policy because a tariff or subsidy changes their sector price and therefore their rents.

Interpretation: The specific-factor structure creates concentrated gains and diffuse losses. This is what makes lobbying plausible: owners of sector-specific factors have a large stake in sectoral protection.

3.3 Trade policy instruments

The home government chooses a vector of trade wedges $\tau = (\tau_1, \dots, \tau_n)$. The domestic price of good i is

$$p_i = \tau_i \pi_i,$$

where π_i is the offshore price. A value $\tau_i > 1$ is an import tariff if the good is imported and an export subsidy if the good is exported. A value $\tau_i < 1$ is an import subsidy or export tax.

Government revenue per capita is

$$r(\tau, \pi) = \sum_i (\tau_i - 1) \pi_i [d_i(\tau_i \pi_i) - X_i(\tau_i \pi_i)].$$

Interpretation: The same wedge notation covers both imports and exports. This is useful because the model studies the whole structure of protection across sectors.

3.4 World market clearing

Let $M_i(p_i)$ denote home net imports and $M_i^*(p_i^*)$ denote foreign net imports. Offshore prices solve

$$M_i(\tau_i \pi_i) + M_i^*(\tau_i^* \pi_i) = 0.$$

Thus π_i is a function of the two countries' policy wedges,

$$\pi_i = \pi_i(\tau_i, \tau_i^*).$$

Interpretation: Each country's policy affects the other through the world price. This is the strategic channel missing from small-country endogenous-protection models.

3.5 Organized lobbies

Let L be the set of politically organized industries at home. A lobby in industry i offers a contribution schedule $C_i(\tau)$ in a trade war or $C_i(\tau, \tau^*)$ in trade talks. Its payoff is

$$V_i = W_i - C_i,$$

where W_i is the gross welfare of the lobby's members.

Interpretation: A contribution schedule is a menu. The lobby does not just offer one donation; it offers a schedule that rewards the government for choosing policies favorable to the lobby.

3.6 Government objective

The home government maximizes

$$G = \sum_{i \in L} C_i + aW.$$

Here W is average national welfare and $a \geq 0$ is the weight politicians attach to voter welfare relative to contributions.

Interpretation: A high a government is more welfare-oriented. A low a government is more politically responsive to contributions. Variation in a is the paper's clean way to study changes in domestic political climate.

4 Models and theoretical specifications

4.1 Equilibrium response in a trade war

For a fixed foreign policy vector τ^* , the home contribution schedules and policy vector form an equilibrium response if the government chooses the policy that maximizes its political objective and no lobby can profitably deviate by offering a different schedule.

Under differentiable contribution schedules, the equilibrium has local truthfulness:

$$\nabla_{\tau} C_i(\tau^0; \tau^*) = \nabla_{\tau} W_i(\tau^0; \tau^*) \quad \text{for all } i \in L.$$

Summing over organized lobbies gives

$$\sum_{i \in L} \nabla_{\tau} W_i(\tau^0; \tau^*) + a \nabla_{\tau} W(\tau^0; \tau^*) = 0.$$

Interpretation: Local truthfulness means a lobby's marginal contribution equals its marginal gross welfare gain from the policy. This converts the contribution game into a tractable first-order condition.

4.2 Trade-war tariff formula

The home trade-war policy can be written schematically as

$$\tau_i - 1 = \underbrace{\text{political support term}}_{\text{organized factor owners vs. average voters}} + \underbrace{\text{terms-of-trade term}}_{\text{large-country optimum tariff motive}}.$$

In the paper's notation, the political term depends on I_{iL} , α_L , a , domestic output X_i , net imports M_i , and domestic price sensitivity. The terms-of-trade term depends on the foreign import-demand or export-supply elasticity.

Interpretation: This decomposition is the core of the trade-war analysis. Political lobbying pushes policy toward organized producer interests; market power in trade pushes policy toward taxing international exchange.

4.3 Organized import and export sectors

- Organized import-competing industries tend to receive protective tariffs.
- Unorganized export industries tend to face export taxes because the government wants to improve the terms of trade.
- Organized export industries generate conflict: the organized sector wants an export subsidy, while aggregate terms-of-trade incentives may favor an export tax.

Interpretation: The signs of policy depend on whether political interests and terms-of-trade motives point in the same direction. For import-competing organized sectors, they reinforce. For export sectors, they can conflict.

4.4 Constant-elasticity special case

The paper uses a constant-elasticity example to compare the political equilibrium with the classic Johnson tariff-war equilibrium. Import demand and export supply have constant elasticities. The Johnson equilibrium is the intersection of benevolent governments' best responses. The political equilibrium shifts these response curves because domestic lobbies change governments' objectives.

Interpretation: The diagrams are not merely illustrations. They show how domestic politics changes the geometry of international policy reactions.

4.5 Political-climate comparative statics

A fall in a means the home government cares less about aggregate welfare relative to contributions. In an import-competing sector, the home political response curve shifts toward higher protection. Foreign policy also adjusts strategically.

Interpretation: A more lobby-sensitive government can sometimes secure better terms of trade because foreign governments expect it to be more aggressive. This creates the paper’s political paradox: selfish politics can sometimes help the home electorate through international strategic effects, even while creating domestic distortions.

4.6 Trade talks and efficient bargaining

In trade talks, lobbies can condition contributions on the policies that emerge from negotiations. The home government maximizes

$$G = \sum_{i \in L} C_i(\tau, \tau^*) + a[W(\tau, \tau^*) + R],$$

while the foreign government maximizes the analogous expression with transfer $-R$.

If governments bargain efficiently, policies maximize a weighted sum of the two political objectives:

$$a^*G + aG^*.$$

Interpretation: Efficiency here is political efficiency from the governments’ perspective, not global social welfare efficiency. Trade talks remove policy choices that leave both governments worse off, but they do not erase domestic interest-group distortions.

4.7 Negotiated policy formula

The negotiated policy wedge is pinned down by relative political strength. In simplified terms,

$$\text{home relative protection} = \text{home political pressure} - \text{foreign political pressure}.$$

In the paper’s formal expression, the policy ratio τ_i/τ_i^* depends on political organization, lobbying population shares, government welfare weights, output and trade exposure, and domestic/foreign price sensitivities.

Interpretation: Unlike the trade-war formula, the negotiated formula excludes pure market-power terms. Efficient bargaining internalizes terms-of-trade externalities. What remains is the cross-country balance of political influence in each sector.

4.8 No-transfer equivalence

The paper shows that the efficient policy ratio is the same even if international transfers are impossible. Governments can mimic transfers by changing both countries’ wedges proportionally while holding τ_i/τ_i^* constant.

Interpretation: This is important because real trade agreements rarely contain arbitrary cash transfers. The model still applies because trade-tax revenue shifts can perform the same surplus-division role.

4.9 Bargaining surplus division

The paper uses a Rubinstein bargaining model with possible breakdown. The trade-war equilibrium is the disagreement point. The division of surplus depends on:

- each government’s fallback political welfare in the trade war;
- each government’s patience in bargaining;
- the chance that negotiations break down before agreement.

Interpretation: The policy ratio is pinned down by efficiency; the split of the surplus is pinned down by bargaining power. Governments with better trade-war outside options or greater patience receive more favorable settlements.

5 Theoretical design and identification logic

5.1 What the model identifies conceptually

The paper identifies the separate roles of four forces:

- domestic interest-group organization;
- politicians' concern for voter welfare versus contributions;
- international terms-of-trade market power;
- bargaining between governments.

Interpretation: These forces are often confounded in reduced-form discussions of trade policy. The model shows which objects enter trade-war outcomes and which enter negotiated outcomes.

5.2 Why local truthfulness matters

Local truthfulness allows the common-agency game to be solved through marginal welfare conditions. Without it, contribution schedules could be arbitrary and policy formulas would be hard to interpret.

Interpretation: Local truthfulness is the bridge between political contributions and policy formulas. It makes contribution schedules reveal the lobbies' marginal willingness to pay for policy changes.

5.3 Why unobserved contribution schedules matter

The model assumes that domestic lobbies' contribution schedules are not observable to the foreign government. If they were observable and binding, lobbies might manipulate them strategically to affect foreign policy responses.

Interpretation: The assumption keeps the delegation game from becoming a commitment game. Domestic lobbies influence their own government, not foreign strategic beliefs directly.

5.4 Why trade talks are not simply free trade

Even if negotiations eliminate terms-of-trade inefficiency, they need not yield free trade. The reason is that domestic governments remain politically motivated, and lobbies still pay for favorable treatment in the negotiated policy vector.

Interpretation: Trade agreements reduce some inefficiencies but leave others. They neutralize international market-power externalities while preserving domestic political distortions.

5.5 What the paper cannot fully prove

- It is a theoretical model rather than an empirical test.
- The set of organized lobbies is exogenous.
- Contribution schedules are implicit and not directly observed.
- The model abstracts from dynamic reputation, repeated trade negotiations, enforcement, uncertainty, and domestic electoral competition in full detail.
- It treats lobbies as industry-level actors and does not model within-industry heterogeneity.

Interpretation: These limitations do not undermine the main conceptual contribution: the paper demonstrates how to combine special-interest politics with strategic international trade policy in a tractable framework.

6 Section-by-section study guide

6.1 Section I: Introduction

The introduction motivates the paper with the observation that trade agreements and trade disputes are shaped by domestic political pressures. The paper positions itself against purely statist or benevolent-government models of trade policy. Its main objective is to merge domestic political economy with international strategic interaction.

6.2 Section II: Model outline and relation to the literature

This section explains how the model relates to earlier endogenous trade-policy work. The key difference is that lobbies offer contribution schedules to influence policy announcements, while governments care about both contributions and voter welfare. The section also explains why the paper moves beyond the earlier small-country setup.

6.3 Section III: The formal model

This section defines consumers, production, specific factors, trade taxes, contribution schedules, government objectives, and world market clearing. It establishes the notation used throughout the paper.

6.4 Section IV: Trade wars

This section studies noncooperative trade policy. It defines equilibrium responses, derives local truthfulness, and obtains the tariff formula with political and terms-of-trade components. It also uses the two figures to show how political response curves differ from Johnson response curves.

6.5 Section V: Trade talks

This section studies cooperative bargaining. The main result is that efficient agreements eliminate pure terms-of-trade motives but retain relative special-interest politics. The section also shows that transfers are not essential and uses a bargaining model to split the surplus.

6.6 Section VI: Conclusions

The conclusion translates the model into testable implications. Protection in trade wars should reflect domestic political organization and trade elasticities. Protection in trade agreements should reflect relative political strength across countries, not pure market power.

7 Tables and figures

The paper has two main theoretical figures and no empirical tables. The two figures are placed on the same row to save space and to make the comparison between trade-war geometry and comparative statics easier to read.

7.1 Figure 1: Trade war equilibrium + Figure 2: Decrease in a^*

Read:

- **Figure 1** plots policy-response curves in (τ, τ^*) space.
- The vertical and horizontal curves BB and B^*B^* represent the Johnson best responses when governments maximize aggregate welfare.
- Point J is the Johnson trade-war equilibrium.
- The curves PP and P^*P^* are political response curves when organized lobbies influence governments.
- Point W is the political trade-war equilibrium.
- In the depicted import-competing case, the political equilibrium lies northeast of J : the home tariff is higher and the foreign export tax is lower, or foreign export subsidy higher.
- **Figure 2** shows how a reduction in the foreign government's welfare weight a^* shifts the foreign political response curve.
- The new equilibrium W' differs from W because the foreign government becomes more responsive to its organized industry.
- The figure illustrates that a domestic political shift in one country changes the other country's strategic policy environment.

Detailed interpretation: Figure 1 compares two equilibrium concepts. The Johnson equilibrium isolates terms-of-trade motives under benevolent governments. The political equilibrium adds domestic lobbying. The home political curve PP shifts away from BB because a home import-competing lobby is willing to pay for protection. The foreign curve P^*P^* changes because foreign organized interests and foreign welfare weight affect the foreign government's response. Point W therefore captures an international policy conflict filtered through domestic politics. Figure 2 then asks what happens when one government's political climate shifts. A fall in a^* makes the foreign government place less weight on aggregate welfare and more relative weight on contributions. This can change both countries' policies through the strategic response curves.

Economic message: The figures show visually that domestic politics is not a purely domestic distortion. A change in one country's lobbying environment changes foreign policy responses, terms of trade, and the distribution of rents across organized groups in both countries.

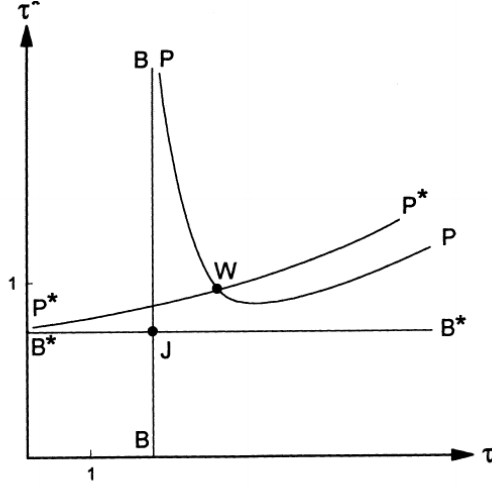


FIG. 1.—Trade war equilibrium

(a) Figure 1: Trade war equilibrium

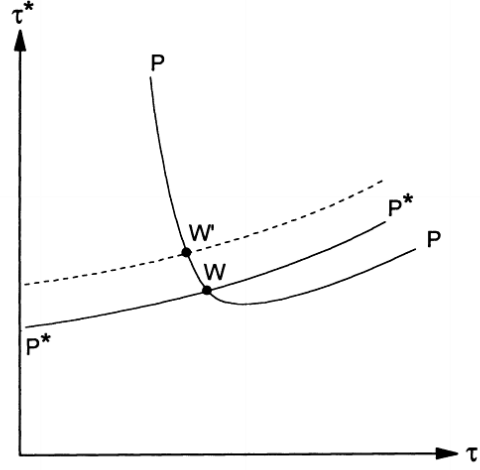


FIG. 2.—Decrease in a^*

(b) Figure 2: Decrease in a^*

7.2 Theoretical result blocks

Result 1: Local truthfulness

In equilibrium, each lobby's contribution schedule is locally truthful:

$$\nabla C_i = \nabla W_i.$$

The marginal promised contribution equals the lobby's marginal gross welfare gain from policy. This allows the contribution game to be summarized by welfare-gradient conditions.

Result 3: Trade talks

Efficient trade negotiations eliminate pure terms-of-trade motives. The negotiated policy wedge depends on relative political pressure at home and abroad.

Result 5: No-transfer equivalence

The efficient policy ratio can be implemented even when direct transfers are not possible. Joint proportional changes in trade wedges can shift government revenue while preserving relative domestic prices.

Result 2: Trade-war decomposition

A noncooperative trade tax has a political-support component and a terms-of-trade component. Organized import-competing sectors obtain protection because political and terms-of-trade incentives reinforce each other.

Result 4: Political power in negotiations

An industry gains under a trade agreement when its domestic lobby is politically stronger than the corresponding foreign lobby. If political strength is symmetric, lobbying pressures cancel and the sector's relative price is closer to free trade.

Result 6: Bargaining surplus division

The division of gains from trade talks depends on trade-war fallback payoffs and patience. A government with a better disagreement payoff or lower bargaining impatience captures more surplus.

8 Detailed result map

- **Model setup.** Two large countries have specific-factor industries, organized lobbies, campaign contributions, and governments that value both contributions and social welfare.
- **Common agency.** Lobbies act as principals and politicians act as common agents. Contribution schedules induce policy choices.

- **Local truthfulness.** Differentiable equilibrium contribution schedules have marginal contributions equal to marginal lobby benefits.
- **Trade war.** Noncooperative policy equals political support plus terms-of-trade incentives.
- **Import-competing sectors.** Organized import-competing industries get tariffs because lobbying and market-power motives point in the same direction.
- **Export sectors.** Export policy is ambiguous when the sector is organized because lobbies want subsidies while terms-of-trade incentives favor export taxes.
- **Comparative statics.** A lower government welfare weight increases political responsiveness and shifts international policy response curves.
- **Trade talks.** Efficient political bargaining eliminates terms-of-trade externalities but preserves relative lobbying effects.
- **Political power.** The negotiated outcome favors the stronger lobby in each paired industry.
- **Surplus split.** Bargaining payoffs depend on disagreement values and patience.

9 Short summary

- Grossman and Helpman combine domestic lobbying and international trade-policy interaction.
- Governments are not benevolent planners; they maximize a weighted sum of campaign contributions and voter welfare.
- Organized industry lobbies offer contribution schedules to influence trade policies.
- The model is a common-agency problem nested inside an international game.
- In a trade war, each country's policy contains a political component and an optimum-tariff component.
- Organized import-competing industries typically receive tariffs.
- Unorganized export sectors tend to face export taxes.
- Organized export sectors create conflict between lobby pressure and terms-of-trade motives.
- A government that becomes more responsive to lobbies shifts the foreign government's strategic response.
- Trade talks are attractive even for self-interested politicians because noncooperative policies impose political costs across countries.
- Efficient trade agreements remove pure terms-of-trade motives.
- Negotiated policies depend on relative political strength of home and foreign lobbies.
- Direct transfers are not necessary because trade-wedge levels can mimic transfers while preserving policy ratios.
- Bargaining surplus depends on trade-war outside options and patience.
- The paper's key empirical implication is that trade agreements should reflect foreign political conditions, not only domestic political variables.

10 Conclusion

10.1 Core conclusion

Grossman and Helpman show that international trade policy cannot be understood by looking only at national welfare or only at domestic lobbying. Trade policy is jointly determined by domestic political

contributions and strategic interaction between governments. In a trade war, governments use policy both to reward domestic lobbies and to manipulate world prices. In trade talks, governments internalize terms-of-trade externalities, but domestic political pressures remain embedded in the negotiated outcome.

10.2 Economic intuition

A tariff affects multiple groups at once. It raises rents for one domestic industry, changes consumer surplus, generates or destroys tax revenue, and shifts the world price. A domestic lobby cares about the first effect most. A welfare-oriented government cares about all domestic effects. A foreign government cares about the world-price spillover and its own lobbies. The equilibrium policy therefore reflects a hierarchy of incentives: lobbies influence governments, governments influence one another, and negotiations transform noncooperative incentives into political bargains.

10.3 Incremental contribution to trade theory

The paper's main advance is to integrate the "protection for sale" logic with large-country strategic trade theory. Johnson-style tariff wars explain how market power produces inefficient trade barriers. Grossman–Helpman-style lobbying explains how interest groups buy protection in small open economies. This paper merges the two, showing that protection in large countries has both a domestic-political and an international-strategic component.

10.4 Incremental contribution to political economy

The paper turns the abstract idea of a two-level game into a formal model. Domestic politics determines each government's international objective, and international bargaining feeds back into domestic political rents. This creates empirical predictions that are richer than standard protection regressions: one must look not only at domestic political organization but also at the political organization of competing foreign industries.

10.5 Policy implications

The analysis suggests that trade agreements are politically valuable because they neutralize cross-country strategic externalities. But they are not necessarily welfare-maximizing because they may encode political bargains between lobbies. A negotiated tariff cut in one sector may reflect foreign lobby strength rather than global efficiency. Therefore, evaluating trade agreements requires attention to political distribution, not just aggregate tariff levels.

10.6 Limitations

The model is deliberately stylized. It does not endogenize which industries organize. It does not model the legal and institutional details of trade agreements. It abstracts from repeated-game enforcement, uncertainty, and voter information. It treats lobbies as sectoral actors and assumes contribution schedules operate as implicit contracts. These simplifications are the cost of obtaining clear analytical results.

10.7 Takeaway

The paper's central lesson is that trade wars and trade talks are both political equilibria. Trade wars combine lobbying and terms-of-trade manipulation; trade talks remove terms-of-trade inefficiency but retain lobbying. International cooperation can therefore reduce some inefficiencies while still reflecting domestic political power.